Gaussian universality of phase-space networks in frustrated and glassy lattice models

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Phase-space networks are connected graphs of many-body configurations as nodes and whose edges are the local moves that connect them, and such definition make an exponentially large phase space into an object that can be inspected. In geometrically frustrated magnets, lattice gases and spin glasses both the degrees of these networks and the spectra of their adjacency matrices have been observed to become Gaussian with system size, but no general derivation has been available. We prove three theorems, that under different conditions, the spectrum of the global adjacency operator, the spectrum inside a conserved-charge sector, and the degree distribution all obey Gaussian statistics. All three are manifestations of the central limit theorem for weakly dependent local contributions with exponentially decaying cumulants resembles Lieb–Robinson bound for many body quantum system. A decimation transformation preserves the limit with rescaled constants, so that phase-space networks sit at a self-similar Gaussian fixed point. These results identify a class of networks, neither Wigner semicircles nor power-law spectrum, and supply the analytical counterpart of the universality observed numerically.

Many-body phase spaces are exponentially large, yet in discrete systems the exact connectivity of the phase space can be enumerated and drawn as a network. Their shape, that is, which configurations are neighbours and whether the graph is a single basin or a landscape of valleys, decides how a material relaxes, ages and yields [1, 2]. For glasses the point is sharp, because the number of configurations grows exponentially with the number of particles and the graph is therefore sampled rather than seen, with colloids, whose micrometre spheres can be followed one by one, only partly lifting that restriction [2, 3].

Geometrically frustrated magnets bring the same problem into a discrete setting, where it can be solved from below. When the competing interactions forbid a unique ground state, the corresponding manifold is macroscopically degenerate [4]. In particular, the triangular-lattice anti-ferromagnet is the archetype [5], and its ground states are in one-to-one correspondence with stacking of cubes [6]; spin ice and the six-vertex model are its higher-dimensional relatives, in which the elementary excitations behave as monopoles [7]. Frustration therefore reproduces macroscopic degeneracy and a rugged landscape to which glass theory attributes while keeping every degree of freedom discrete, and buckled colloidal crystals realise the physics in the laboratory [8].

Discreteness turns a phase space into a graph, and the graph can be enumerated. Continuous landscapes are normally coarse-grained into networks of basins first [9]; for a discrete system no such coarse-graining is needed. Taking each configuration as a node and each zero-energy local rearrangement as an edge, Han and co-workers constructed the exact phase-space networks of geometrically frustrated systems [10], the six-vertex model [11], spin glasses [12] and lattice gases [13]. In all of them a signature appeared that neither random-matrix theory nor the theory of complex networks anticipates: as the system grows, both the degree distribution and the adjacency spectrum become Gaussian, while the network itself remains self-similar under coarse-graining [13]. Neither Wigner ensembles [14] nor the semicircle law of sparse random graphs [15] yields a Gaussian density of states, so the observation pointed to a mechanism specific to these systems.

Here we prove three such statements and show that they are one statement. Each concerns a different network observable, each holds under a different condition on the local moves, and each asserts that a normalised sum of many weakly dependent local contributions has a Gaussian law in the thermodynamic limit, which is the central limit theorem (CLT), the dependence being controlled by cumulants that decay exponentially with the diameter of the supports involved (Theorem 1). A conserved charge commuting with the adjacency operator splits the problem into sectors and the same argument applies sector by sector (Theorem 2). Replacing every local operator by its row-sum function, that is, replacing the trace by a uniform average over basis states, turns the identical computation into a statement about the degrees of the network (Proposition 1). The three Gaussian distributions are therefore the density of states of the global operator, the conditional density of states inside a charge sector, and the degree distribution. Each is a normal law $N(0, \sigma^2)$, that is, a density proportional to $e^{-x^2/2\sigma^2}$ whose only parameter is the variance, fixed here by two-point correlations, and they differ only in which quantity is summed and in which condition renders the summands weakly dependent.

Three Gaussian limits.— The results are stated here and proved in the appendices; the hypotheses are collected in Appendix and Appendix .

Theorem 1 (Gaussian limit for non-commuting sums)**.** *Under assumptions (A1)–(A3), the empirical spectral*

distribution of

$$\frac{A_N}{\sqrt{N}}$$

converges weakly, as $N \to \infty$ with $L_1, \ldots, L_d \to \infty$, to the centred Gaussian law with variance

$$\sigma^2 := \lim_{N\to\infty} \sum_{j\in\Lambda} \kappa_2(X_0, X_j) = \lim_{N\to\infty} \sum_{j\in\Lambda} \tau(X_0 X_j), \quad (1)$$

which is finite and non-negative. More precisely, for every integer $m \geq 0$,

$$\lim_{N\to\infty} \tau\left[\left(\frac{A_N}{\sqrt{N}}\right)^m\right] = \begin{cases} (m-1)!!\,\sigma^m, & m \text{ even}, \\ 0, & m \text{ odd}. \end{cases}$$

Consequently,

$$\boxed{\rho_{\frac{A_N}{\sqrt{N}}}(\lambda) \xrightarrow[N\to\infty]{} \mathcal{N}(0, \sigma^2)} \quad (2)$$

in the weak topology, provided $\sigma^2 > 0$.

Theorem 2 (Gaussian limit under a conservation law)**.** *Suppose that (A1) and (C0)–(C2) hold. For each sector q with $p_q > 0$, define the (finite) sector variance*

$$\sigma_q^2 := \sum_{j\in\Lambda} \kappa_2^{(q)}(X_0, X_j),$$

Then, for every fixed q, the conditional empirical spectral density of $A_N/\sqrt{N}$ restricted to $\mathcal{H}_q$ converges weakly to the centred Gaussian law:

$$\rho_{\frac{A_N}{\sqrt{N}}\,|\,Q=q}(\lambda) \xrightarrow[N\to\infty]{} \mathcal{N}(0, \sigma_q^2)$$

If, in addition, the concentration condition (C3) holds, then the unconditional empirical spectral density of $A_N/\sqrt{N}$ converges weakly to

$$\rho_{\frac{A_N}{\sqrt{N}}}(\lambda) \xrightarrow[N\to\infty]{} \mathcal{N}(0, \sigma^2),$$

In this case the result reduces exactly to that of Theorem 1 and Q is said to be trivial.

Proposition 1 (Gaussian connectivity fluctuations)**.** *Assume (A1)-(A5), then the empirical distribution of*

$$\frac{1}{\sqrt{N}} \sum_{i\in\Lambda} T_i(x)$$

with respect to the uniform measure on the computational basis converges weakly, as $N \to \infty$, to a centred Gaussian law with variance

$$\sigma_{\text{row}}^2 := \lim_{N\to\infty} \sum_{j\in\Lambda} \mathbb{E}_x[T_0(x) T_j(x)].$$

If, in addition, a conserved charge Q is present and the sector assumptions (C0)-(C2)&(C4) hold, then the same conclusion holds in each sector after conditioning, and the global distribution is again Gaussian with variance $\mathbb{E}_Q[\sigma^2_{q,\text{row}}]$, provided the sector variances concentrate as in (C3).

One mechanism, three statements.— Theorem 1 is the general statement: it assumes only that the moves are Hermitian and translation covariant, traceless, and exponentially clustering in the sense of (A1)–(A3) of Appendix , and it produces the first Gaussian, the density of states of $A_N/\sqrt{N}$. Theorem 2 follows from it by restriction: a conserved charge block-diagonalises A_N, so inside each sector the hypotheses of Theorem 1 hold with the sector trace τ_q, which yields the sector-wise Gaussian $N(0, \sigma_q^2)$; the mixture over sectors is Gaussian only when the sector variances concentrate, and the unconditional statement is exactly that condition (C3) of Appendix . Proposition 1 is Theorem 1 under the substitution $X_i \mapsto T_i(x)$, $\tau \mapsto \mathbb{E}_x$ and $(A3) \mapsto (A5)$: every step of the moment computation, namely the cumulant expansion, the counting of partitions and the pairing dominance, is unchanged by the substitution, so the degrees of the network obey the same CLT and the third Gaussian follows.

The Gaussian limit.— To begin with, interaction on high dimensional lattice are introduced formally as local notation of operators. Let Λ be a periodic lattice of N sites, X_0 a Hermitian operator and X_i its translates, and let $A_N = \sum_{i\in\Lambda} X_i$ be the adjacency operator, whose trace-normalised density of states is ρ_{A_N}. There are two sufficient hypotheses that fix the statistics: the moves are translation covariant and traceless, and their cumulated traces decay as $\kappa_m \leq C_m e^{-\mu \operatorname{diam}}$. Then

$$\begin{aligned} \rho_{A_N/\sqrt{N}}(\lambda) &\xrightarrow[N\to\infty]{} \mathcal{N}(0, \sigma^2), \\ \sigma^2 &= \lim_{N\to\infty} \sum_{j\in\Lambda} \kappa_2(X_0, X_j), \end{aligned} \quad (3)$$

in the weak topology, and the convergence is accompanied by the moment statement $\tau[(A_N/\sqrt{N})^m] \to (m-1)!!\,\sigma^m$ for even m and to zero for odd m. The proof (SI Appendix) is a moment computation in which only pairings survive. In a pairing scheme, a partition of the m factors of the trace into b blocks contributes at order $N^{b-m/2}$, so that partitions with $b < m/2$ vanish and the surviving matchings reproduce the Gaussian moments. The variance σ^2 is the sum of two-point correlations, which is finite by the decay hypothesis and model specific, unlike the Gaussian distribution with proper normalization. The same decay hypothesis fixes the row sums, so the degrees of the network become Gaussian at the same time as the spectrum for the same reason (Fig. S3 and SI Appendix).

The exactly solvable core.— The limit has an exactly solvable representative, and it is the reason the phe-

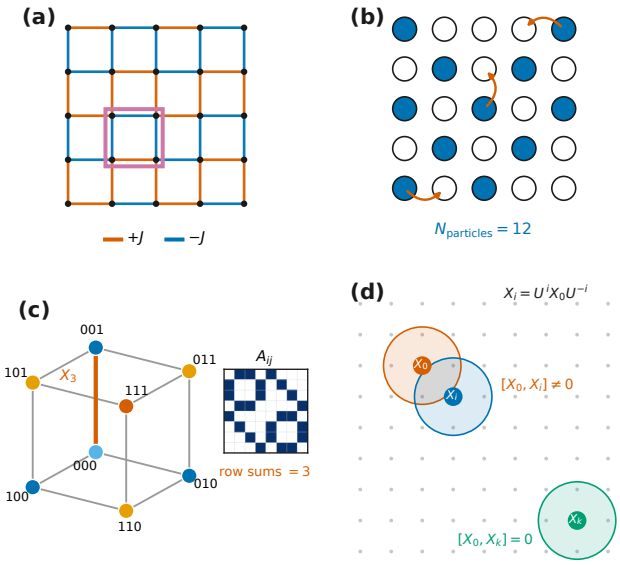


FIG. 1. Frustrated discrete systems and their phase-space networks. (a) A square-lattice $\pm J$ spin glass; the highlighted plaquette is frustrated, since no spin assignment satisfies all four of its bonds. (b) A lattice gas whose elementary move exchanges an occupied and an empty site, conserving the particle number. (c) The configuration graph for $N = 3$ bits: a hypercube whose vertices are configurations and whose edges are single spin flips, together with its adjacency matrix. (d) Translation-covariant local operators $X_i = U^i X_0 U^{-i}$ on a periodic lattice; the commutator $[X_i, X_j]$ is controlled by the overlap of the supports, and vanishes when the supports are disjoint.

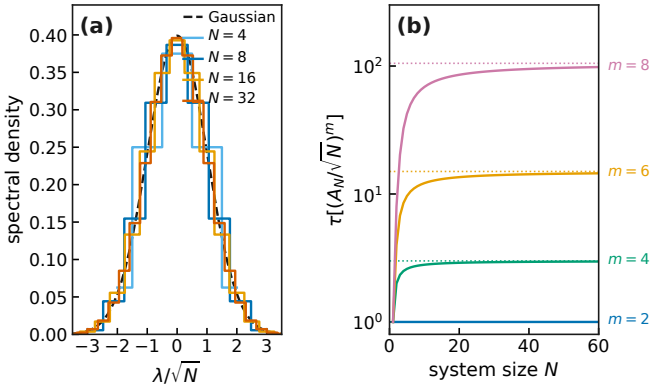


FIG. 2. The exactly solvable limit. (a) Exact hypercube spectra for $N = 4, 8, 16, 32$, approaching the Gaussian law of Eq. (3). (b) Even moments of the normalised adjacency operator converging to $(m-1)!!$. Here the Gaussian limit is the classical central limit theorem applied to a sum of independent Rademacher variables.

nomenon was noticed in the first place. For a (not necessary two dimensional) periodic boundary lattice gas, whose vertices are configurations and edges are independent single spin flips, the configuration graph is a hypercube. In this picture, every vertex has N neighbours whose local moves commute, and the adjacency operator is diagonalised at once. Its eigenvalues are sums of N independent Rademacher variables, so the classical central limit theorem gives a Gaussian density of states, with binomial degeneracies controlling the approach for finite N (Fig. 2). This case is a warm-up reference point where everything that follows asks how much of it survives once the moves cease to commute and the row sums cease to be constant. The elementary derivation, including the degeneracy formulae and the uniform error term, is reproduced in SI Appendix .

Locality is enough.— To generalize the hypercube, the quasi locality hypothesis is replaced by commutativity and constant degree. The nested commutators of the local operators decay are exponentially in the diameter of the supports (SI Appendix), and for finite-range moves the commutator vanishes identically as soon as the supports disjoint. Therefore the bound holds with constants independent of the lattice size (Fig. 1d). Its operative form in the proof is the exponential clustering of cumulants, which for finite-range operators follows from quasi-locality; SI Appendix states the hypothesis, discusses why it is sufficient rather than necessary, and calibrates its decay constants on a toy model (Fig. S1). Conclusively, inn models with all-to-all coupling the clustering fails, and so does the Gaussian limit (SI Appendix), and thus The hypothesis is not vacuous.

A conservation law splits the spectrum.— For a lattice gas with a conserved charge Q, e.g. particle number, energy or magnetisation, the real phase space also requires further constraints. In the language of operators where Q is formally an operator that commutes with A_N makes A_N block diagonal, the full density of states becomes a mixture of the sector densities weighted by the sector dimensions (Fig. 3a). Each sector inherits the Gaussian limit, with its own variance σ_q^2, so the full spectrum is a mixture of Gaussians and is itself Gaussian only when the variances concentrate (Fig. 3b). This is not a technicality. A mixture of centred Gaussians is Gaussian if and only if the variance is almost surely constant, which can be shown by Jensen's inequality, so any non-trivial dependence of the sector variance on the charge necessarily produces a non-Gaussian limit (SI Appendix). The concentration condition is therefore forced by the data rather than imposed by us, and it is the precise sense in which a conservation law may be called trivial.

Worked examples.— Two typical models, both analysed numerically in the works cited above, show how the hypotheses are checked in practice. In a $\pm J$ spin glass on a square lattice, frustration produces a macroscopic ground-state degeneracy and the elementary moves are

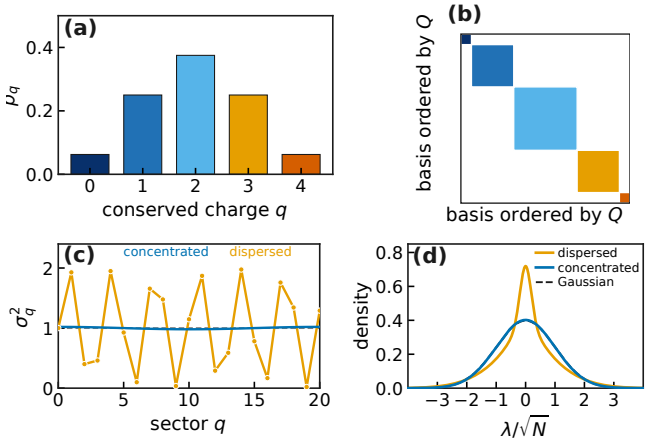


FIG. 3. A conserved charge. (a) For a small system ($N = 4$) with conserved particle number $Q = \sum_i n_i$, the Hilbert space splits into sectors $\mathcal{H}_q$ of weight $p_q = \dim \mathcal{H}_q / 2^N$. (b) In the basis ordered by Q the adjacency operator is block diagonal, $A_N = \bigoplus_q A_N^{(q)}$. (c) The sector variances σ_q^2, shown for two cases: dispersed across sectors (orange) and concentrated on a common value (blue). (d) The full spectrum is the mixture $\rho = \sum_q p_q \rho_q$. A dispersed σ_q^2 leaves the mixture visibly non-Gaussian even though every sector is asymptotically Gaussian, whereas a concentrated σ_q^2 collapses it onto a single Gaussian; this is the concentration condition of SI Appendix .

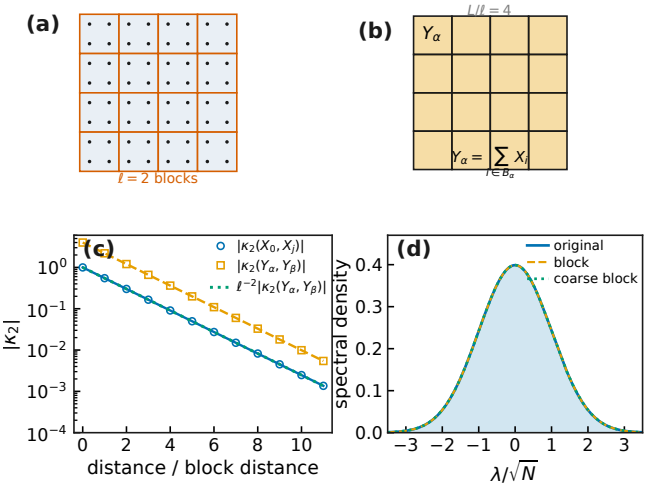


FIG. 4. Decimation and the Gaussian fixed point. (a) The lattice is partitioned into blocks of linear size ℓ. (b) Each block is replaced by the sum of its local operators, giving an approximately translation-invariant family on the coarse lattice. (c) Iterating the map produces the next-level block variables. (d) The exponential decay of the cumulants is preserved with rescaled constants, so that the normalised density of states is invariant and the network is asymptotically self-similar.

single spin flips; each X_i therefore has finite support, the nested commutators of the quasi-locality hypothesis vanish identically for distant sites, and the exponential clustering of cumulants follows from the product structure of the maximally mixed state, so the Gaussian limit applies without any estimate. A lattice gas with conserved particle number is the same construction one level up. Its moves exchange an occupied and an empty site, hence they are local, and the conserved charge merely splits the operator into sectors, each of which inherits the limit. In both cases the hypotheses are verified by inspection rather than by calculation, which is the practical content of the theorem; the details, together with the remark on non-Bravais lattices, are collected in SI Appendix .

Coarse-graining and self-similarity.— The limit is stable under coarse-graining, and this is what connects the construction to glass theory. Group the lattice into blocks and replace each block by the sum of the moves it contains; the block sums are again translation covariant, and because a block sum is a finite sum the exponential clustering survives with rescaled constants (Fig. 4). The map is thus a renormalisation-group transformation with a Gaussian fixed point [16, 17]. Therefore the phase-space network is asymptotically self-similar, given that iteration leaves the density of states invariant up to rescaling. The transformation is not exact at finite block size, since the non-commutativity of the moves generates corrections from nested commutators. Those corrections are exponentially small in the separation of the blocks, so the fixed point is asymptotic rather than exact (SI Appendix).

The self-similarity deserves to be separated from a stronger claim. Fractal landscapes are a postulate of several theories of glasses [18, 19], self-similar phase-space networks have been reported for particular frustrated models [13], and fractality is in any case a generic feature of real complex networks [20]. What the present result adds is a derivation that does not depend on the model. Every translation-covariant, exponentially clustering family sits at the fixed point, so self-similarity follows from locality rather than from the details of a lattice. In this sense the phase-space network supplies a finite-dimensional, explicitly computable example of the hierarchical landscapes that glass theory has had to assume.

Consequences.— Three consequences seem worth stating. First, the networks in question form a class of their own. Dense random matrices give the Wigner semicircle [14] and sparse random graphs give a semicircle with localised edges [15]; the networks discussed here give Gaussians, with the variance as the only parameter that survives the thermodynamic limit. Any discrete system whose moves are local and few-body will show the same behaviour, which makes Gaussianity a diagnostic of locality rather than of a particular Hamiltonian.

Second, the mechanism is what makes the phenomenon

experimentally accessible. Frustrated colloidal crystals are a physical realisation of the triangular antiferromagnet [8], and phason flips in quasi-crystals are the same kind of zero-energy local move; wherever the configuration space can be enumerated, the Gaussian signature is the observable counterpart of the theorem, and wherever it cannot, the theorem is a prediction about the limit that enumeration cannot reach.

Third, the framework separates what is universal from what is not. The variance, the sector structure and the position of the spectral weight are model-specific; the Gaussian shape is not. This is the sense in which the phase-space network can be useful to glass theory where it is a quantity measured in a small system and then extrapolated analytically, which is precisely what sampling cannot do.

Outlook.— Several questions remain open. At finite temperature the moves acquire Boltzmann weights, the adjacency matrix becomes non-symmetric, and the spectrum moves into the complex plane, where the natural limits are circular rather than Gaussian; a crossover between the two laws as the temperature is raised is the obvious conjecture. Whether the machinery survives long-range interactions, and how the entropy of a phase space should be defined when the landscape is self-similar and the entropy therefore not additive [18], are further questions that the present framework makes precise rather than answers. All hypotheses, proofs and technical developments are collected in SI Appendices –.

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List of symbols

Mathematics symbols

$\mathbb{Z}$:
the ring of integers
$\mathbb{Z}_L$:
the cyclic group of integers modulo L (used for lattice directions)
$\mathbb{C}$:
the field of complex numbers
$\mathbb{E}(S)$:
the probabilistic expectation of S
$\mathcal{N}(\mu, \sigma^2)$:
the Gaussian distribution of mean μ and variance σ^2
$\mathcal{H}$:
the Hilbert space of a system
$\mathcal{B}(\mathcal{H})$:
the space of bounded linear operators on $\mathcal{H}$
$\|\cdot\|$:
the operator norm on $\mathcal{B}(\mathcal{H})$
$\oplus$:
the direct sum of Hilbert spaces or operators
$\otimes$:
the tensor product of Hilbert spaces or operators
$:=$:
the left hand side is defined to be the right hand side

Physical symbols

N:
total number of lattice sites (bits)
A:
adjacency matrix of the configuration graph (as a global operator)
X_j:
local operator acting on site j
U, V:
unitary shift operators on the two-dimensional torus
U_k:
unitary shift operator along lattice direction k
κ_m:
m-th order tracial cumulant
τ:
normalized trace on $\mathcal{B}(\mathcal{H})$
Λ:
finite periodic lattice
S_i:
row-sum function associated with site i
T_i:
centred row-sum function $T_i = S_i - \mathbb{E}[S_i]$
λ:
eigenvalue of the adjacency matrix
ρ_A:
empirical spectral density of A
σ^2:
limiting variance of the Gaussian law
Q:
conserved charge operator
q:
eigenvalue (as a sector label) of Q

The exactly solvable hypercube model

This appendix reproduces the derivation summarised in the main text; the corresponding figures are Figs. 1 and 2.

To establish a systematic framework for understanding the universality observed in the spectral properties of phase-space connectivity matrices, it is natural to begin with a model with almost the highest possible symmetry. Among all lattice-gas configurations on a finite torus, the graph whose vertices represent all 2^N occupation-number configurations and whose edges connect configurations differing by a single spin flip forms a hypercube. This hypercube geometry is distinguished not only by its exact solvability but also by the fact that every vertex has precisely N neighbours. Consequently, the adjacency matrix of this configuration graph has constant row sum and column sum, a property that is intimately related to the commutativity of the local spin-flip operators X_j.

Model setup on a 2D torus

Consider a lattice gas system defined on a two-dimensional periodic lattice (torus)

$$\Lambda = \mathbb{Z}_{L_1} \times \mathbb{Z}_{L_2}, \qquad N := |\Lambda| = L_1 L_2,$$

where N denotes the total number of classical bits (representing spins or occupied sites). The full configuration (Hilbert) space is

$$\mathcal{H} = \bigotimes_{(i,j)\in\Lambda} \mathbb{C}^2, \qquad \dim \mathcal{H} = 2^N.$$

To incorporate the two independent directions of translational symmetry, we introduce two unitary shift operators U and V acting on $\mathcal{H}$:

$$U\,|s_{i,j}\rangle = |s_{i-1,j}\rangle, \qquad V\,|s_{i,j}\rangle = |s_{i,j-1}\rangle,$$

where $s_{i,j} \in \{0, 1\}$ denotes the occupation number at vertex (i, j). These shifts satisfy the exact periodic boundary conditions

$$U^{L_1} = I, \qquad V^{L_2} = I, \qquad [U, V] = 0.$$

Let the local interaction operator at the origin be the Pauli matrix σ_x acting solely on that site:

$$X_{0,0} := \sigma_x^{(0,0)} \otimes I_{\Lambda\setminus\{(0,0)\}}.$$

Using the translational symmetry, we define the interaction at an arbitrary vertex (i, j) as

$$X_{i,j} := U^i V^j X_{0,0} V^{-j} U^{-i}, \qquad (i, j) \in \Lambda.$$

Since the shifts merely permute the tensor factors, this operator is precisely the local σ_x acting on vertex (i, j). The adjacency matrix is defined as the sum over all local operators:

$$A := \sum_{(i,j)\in\Lambda} X_{i,j}. \tag{4}$$

Exact diagonalization: the hypercube limit

We first consider the simplest case where every vertex is completely independent. Since each $X_{i,j}$ acts nontrivially on a distinct tensor factor, they all commute:

$$[X_{i,j}, X_{i',j'}] = 0, \qquad \forall (i, j), (i', j') \in \Lambda.$$

Thus, A is exactly diagonalized in the joint eigenbasis of all $X_{i,j}$. Let $X_{i,j}|\{\sigma\}\rangle = \sigma_{i,j}|\{\sigma\}\rangle, \sigma_{i,j} \in \{\pm 1\}$. Then the eigenvalue corresponding to the configuration $\{\sigma\} \in \{\pm 1\}^\Lambda$ is

$$\lambda(\{\sigma\}) = \sum_{(i,j)\in\Lambda} \sigma_{i,j} = \sum_{n=1}^{N} \sigma_n,$$

where we relabelled the N sites by a single index n for notational convenience. The degeneracy of an eigenvalue $\lambda = N - 2k$ (with k being the number of -1's) is given by the binomial coefficient

$$\deg(\lambda) = \binom{N}{k}, \qquad k = \frac{N - \lambda}{2}. \tag{5}$$

The empirical spectral density is the distribution of the sum of N Rademacher independent variables:

$$\rho_A(\lambda) = \frac{1}{2^N} \sum_{\{\sigma\}} \delta\left(\lambda - \sum_{n=1}^{N} \sigma_n\right).$$

By the classical central limit theorem (CLT), as $N \to \infty$ (with $L_1, L_2 \to \infty$),

$$\boxed{\rho_A(\lambda) = \frac{1}{\sqrt{2\pi N}} \exp\left(-\frac{\lambda^2}{2N}\right) + o(1),} \tag{6}$$

where the error term is uniform in λ. This is the rigorous Gaussian limit for the hypercube graph.

In this extremally symmetric setting, all local operators X_j commute with one another, so the global adjacency operator A is diagonalised in a common eigenbasis and its spectral density is exactly that of a sum of N independent Rademacher variables. The central limit theorem then immediately yields a Gaussian limiting density of states. While this hypercube limit serves as an important reference point, the real challenge and the main focus of this work lies in understanding how the Gaussian universality survives when we perturb away from this highly symmetric structure. The subsequent sections develop a general framework in which the local operators need not commute, and the constancy of the row sums is replaced by the much weaker condition of exponential decay of correlations. The hypercube model thus provides both a pedagogical starting point and a sharp limiting case against which more general results can be tested.

The general framework and its hypotheses

Having established the Gaussian limit for the highly symmetric hypercube model, we now turn to a more general framework that captures the universal behaviour of a broader class of lattice systems. Consider a set of bits placed on a high-dimensional periodic lattice, where the interaction strength between two sites is governed by their spatial distance. To encode the locality of interactions, we introduce local operators X_i associated with each lattice site and impose periodic boundary conditions through unitary shift operators. The concept of Lieb–Robinson bounds, borrowed from the theory of quantum many-body systems, provides a natural tool to quantify the spatial decay of correlations. Our central goal is to demonstrate that, under sufficiently strong locality conditions, the empirical spectral distribution of the global adjacency operator $A_N = \sum_i X_i$ again converges to a Gaussian law. To this end, we propose a set of sufficient conditions in terms of exponential decay of trace cumulants to guarantee a Gaussian limit. We emphasize that these conditions are by no means necessary; there may exist weaker assumptions that yield the same conclusion. Conversely, for certain physical models, verifying the Gaussian limit may require imposing even stronger decay properties than those stated here.

Generalized local operators

Let the lattice be $\Lambda = \mathbb{Z}_{L_1} \times \cdots \times \mathbb{Z}_{L_d}$ with d independent directions, and set

$$N := |\Lambda| = L_1 L_2 \cdots L_d.$$

Let $\mathcal{H}$ be the associated 2^N-dimensional Hilbert space. We fix a Hermitian operator $X_0 \in \mathcal{B}(\mathcal{H})$, not necessarily

acting only on the origin, and define its translates by

$$X_i := U_1^{i_1} U_2^{i_2} \cdots U_d^{i_d}\, X_0\, U_d^{-i_d} \cdots U_2^{-i_2} U_1^{-i_1}, \quad i = (i_1, \ldots, i_d) \in \Lambda,$$

where $U_1, \ldots, U_d$ are unitary shift operators satisfying $U_k^{L_k} = I$ for each k and $[U_k, U_l] = 0$ for all k, l. Thus the family $\{X_i\}$ is translation covariant by construction. We define the global operator

$$A_N := \sum_{i \in \Lambda} X_i. \tag{7}$$

Because the X_i need not commute, A_N is generally not diagonalizable in a common eigenbasis. Nevertheless, under suitable decay conditions its spectral distribution still converges to a Gaussian law in the thermodynamic limit. Let

$$\tau(\cdot) := \frac{1}{2^N} \operatorname{tr}(\cdot)$$

be the normalized trace on $\mathcal{B}(\mathcal{H})$. We will prove convergence of the empirical spectral measure of $A_N/\sqrt{N}$ by the method of moments.

Assumptions

We impose the following conditions.

(A1) *Hermiticity and translation covariance.* Each X_i is Hermitian, $X_i = X_i^\dagger$, and the family is translation covariant:

$$X_i = U_1^{i_1} \cdots U_d^{i_d}\, X_0\, U_d^{-i_d} \cdots U_1^{-i_1}.$$

(A2) *Zero mean.*

$$\tau(X_i) = 0 \qquad \forall\, i \in \Lambda.$$

(A3) *Exponential decay of cumulated traces.* For each ordered m-tuple $(i_1, \ldots, i_m)$, define the classical cumulant $\kappa_m(X_{i_1}, \ldots, X_{i_m})$ (see Appendix) recursively by

$$\tau(X_{i_1} \cdots X_{i_m}) = \sum_{\pi \in \Pi_m} \prod_{B \in \pi} \kappa_{|B|}(X_{i_a} : a \in B),$$

where Π_m is the set of all partitions of $\{1, \ldots, m\}$. This recursion defines κ_m uniquely without commutativity assumptions. We assume that there exist constants $C_m > 0$ and $\mu > 0$, independent of N, such that for every $m \geq 2$ and all ordered tuples $(i_1, \ldots, i_m)$,

$$|\kappa_m(X_{i_1}, \ldots, X_{i_m})| \leq C_m\, e^{-\mu\, \operatorname{diam}(i_1, \ldots, i_m)}.$$

(A4) *Quasi-locality (optional).* There exist constants $C_k' > 0$ and $\mu' > 0$ such that for all $k \geq 2$,

$$\big\| [X_{i_1}, [X_{i_2}, \cdots [X_{i_{k-1}}, X_{i_k}] \cdots]] \big\| \leq C_k'\, e^{-\mu'\, \operatorname{diam}(i_1, \ldots, i_k)}.$$

(A5) *Exponential decay of classical cumulants of centred row-sum functions.* For each site $i \in \Lambda$, define the classical function

$$S_i(x) := \sum_{y \in \{0,1\}^N} \langle x | X_i | y \rangle,$$

and let

$$T_i(x) := S_i(x) - \mathbb{E}_x[S_i],$$

where $\mathbb{E}_x$ denotes the uniform average over the computational basis. Define the classical cumulants $\kappa_m^{\mathrm{cl}}(T_{i_1}, \ldots, T_{i_m})$ with respect to $\mathbb{E}_x$ by the same recursion as in (A3), with the trace τ replaced by the expectation $\mathbb{E}_x$. We assume that there exist constants $C_m > 0$ and $\mu > 0$, independent of N, such that for all $m \geq 2$ and all ordered tuples $(i_1, \ldots, i_m)$,

$$\big|\kappa_m^{\mathrm{cl}}(T_{i_1}, \ldots, T_{i_m})\big| \leq C_m\, e^{-\mu\, \operatorname{diam}(i_1, \ldots, i_m)}.$$

Remark 1. Assumptions (A1) and (A2) concern the basic algebraic and statistical properties of the local operators. It is natural to assume Hermitian X_i since the definition of a spin flip allows the identification of 0 and 1 up to a sign, the symmetry of the configuration space supports a Hermitian representation of the local operation. Therefore this in turn ensures that all eigenvalues of the global adjacency operator are real. By construction, the adjacency matrix of the configuration graph has zero diagonal and is symmetric; the operator X_i represents the effective legitimate operation centred at site i. When the translational symmetry operations are properly defined, traceless A_0 (i.e. $\tau(X_0) = 0$) gives rise to the zero-mean condition (A2), which guarantees that the average contribution of each site vanishes.

Assumptions (A3) and (A4) are closely related. Assumption (A3) requires that all connected correlations of the local operators with respect to the maximally mixed state decay exponentially with the spatial diameter. This exponential decay ensures that distant operators are effectively independent, so that the sum A_N behaves like a sum of weakly dependent random variables. Assumption (A4) is not directly used in the moment computation, but it provides a sufficient condition based on the decay of multiple commutators. Indeed, for operators generated by finite-depth circuits or by short-time evolution under a local Hamiltonian, Lieb-Robinson bounds imply the required exponential decay of commutators; for strictly local X_0 with finite support, commutators vanish

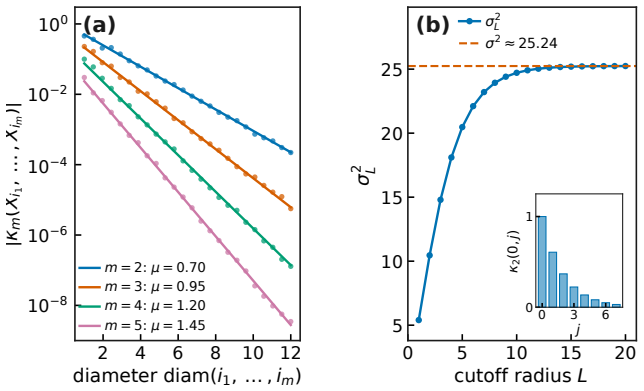


FIG. S1. Locality input and convergence of the limiting variance, illustrated with a toy model. **Left:** exponential decay of cumulated traces, for $m = 2, 3, 4, 5$. The toy-model parameters are $(C_2, C_3, C_4, C_5) = (1.00, 0.55, 0.25, 0.10)$ and $(\mu_2, \mu_3, \mu_4, \mu_5) = (0.70, 0.95, 1.20, 1.45)$; the spatial diameter is sampled as $d = 1, \ldots, 12$, and multiplicative log-normal noise with standard deviation 0.11 mimics finite-size fluctuations. **Right:** partial sum $\sigma_L^2 = \sum_{|j| \le L} \kappa_2(X_0, X_j)$ for a short-range second cumulant $\kappa_2(0, j) = Ae^{-|j|/\xi}$ with $A = 1.0$ and correlation length $\xi = 2.0$, computed on a two-dimensional square cutoff of radius $L = 1, \ldots, 20$. The red dashed line is the limiting variance $\sigma^2 = \lim_{L\to\infty} \sigma_L^2$. The inset shows the short-range behaviour of $\kappa_2(0, j)$ for $j = 0, \ldots, 7$. This toy model illustrates assumption (A3) and the finite limiting variance entering Theorem 1.

identically for disjoint supports, so (A4) holds and (A3) follows. The general mechanism behind this implication is that connected correlations of order higher than two decay exponentially outside the Lieb–Robinson cone, as shown in [21].

The classical cumulant decay in assumption (A5) is the counterpart of (A3) for the row-sum functions T_i. While (A3) controls the cumulants of the operator entries, (A5) imposes the same exponential decay on the classical cumulants of the centred row sums. This is precisely the condition that generalises the constant row sum of the hypercube model, where every row sum is exactly N and here the row sums become random variables with exponentially decaying correlations. We will show in Proposition 1 that, under (A5), the empirical distribution of the normalised row sums also converges to a Gaussian law, thereby establishing a classical analogue of the main theorem.

The theorem itself is stated in the main text; its proof occupies Appendix . The toy model that calibrates the decay constants of the cumulants, and the convergence of the partial sums of the limiting variance, are shown in Fig. S1.

Its hypotheses are (A1)–(A3) above.

The cumulant expansion and the proof of Theorem 1

The two subsections below supply, respectively, the cumulant calculus and the proof of the limit theorem. The argument is a direct moment computation rather than an appeal to an abstract limit theorem; the prototype of the phenomenon, namely that exponential mixing forces Gaussian moments, is the classical limit theorem for stationary mixing random fields [22], and the bookkeeping of cumulant orders by which the non-Gaussian terms are discarded is the classical one [23].

Explicit cumulant formula and examples

For completeness, we provide the closed-form expression for the cumulated traces used in Assumption (A3). Let $X_1, \ldots, X_m$ be operators on $\mathcal{H}$. The m-th order cumulant is defined by the Möbius inversion on the partition lattice:

$$\kappa_m(X_1, \ldots, X_m) = \sum_{\pi \in \Pi_m} (-1)^{|\pi|-1} (|\pi|-1)! \prod_{B \in \pi} \tau\left(\prod_{a \in B} X_a\right),$$

where Π_m is the set of all partitions of $\{1, \ldots, m\}$, $|\pi|$ is the number of blocks in π, and for each block $B = \{a_1 < \cdots < a_k\}$, the product $\prod_{a \in B} X_a$ is taken in increasing index order. This formula is valid without any commutativity assumption; it is the standard partition-lattice Möbius inversion of the theory of non-commutative cumulants [24, 25]. What the present work adds is the passage to the Gaussian limit, which requires translation covariance and exponential clustering rather than finite-dimensional matrix asymptotics.

Example 1 (m=2). For two operators X, Y,

$$\kappa_2(X, Y) = \tau(XY) - \tau(X)\tau(Y),$$

which is the usual connected two-point function.

Example 2 (m=3). For three operators X, Y, Z,

$$\kappa_3(X, Y, Z) = \tau(XYZ) - \tau(X)\tau(YZ) - \tau(Y)\tau(XZ) - \tau(Z)\tau(XY) + 2\,\tau(X)\tau(Y)\tau(Z).$$

If all operators have zero mean (as in Assumption (A2)), this simplifies to

$$\kappa_3(X, Y, Z) = \tau(XYZ) - \tau(X)\tau(YZ) - \tau(Y)\tau(XZ) - \tau(Z)\tau(XY).$$

Proof of Theorem 1

Proof. We use the method of moments, in the pairing (Wick) form familiar from Wigner's moment method [14, 26]. For each integer $m \ge 1$, write

$$M_m(N) := \tau\left[\left(\frac{A_N}{\sqrt{N}}\right)^m\right] = N^{-m/2} \sum_{i_1, \ldots, i_m \in \Lambda} \tau(X_{i_1} \cdots X_{i_m}).$$

Using the cumulant expansion,

$$\tau(X_{i_1}\cdots X_{i_m}) = \sum_{\pi\in\Pi_m}\prod_{B\in\pi}\kappa_{|B|}(X_{i_a}: a\in B),$$

we obtain

$$M_m(N) = N^{-m/2}\sum_{\pi\in\Pi_m}\prod_{B\in\pi}\left[\sum_{\{i_a: a\in B\}}\kappa_{|B|}(X_{i_a}: a\in B)\right],$$

where the outer sum is over all partitions π of $\{1,\dots,m\}$.

By assumption (A2), $\kappa_1(X_i)=\tau(X_i)=0$, so no block of size 1 can occur. Hence every block $B\in\pi$ has size $|B|\geq 2$. Let $b=|\pi|$ be the number of blocks. Then

$$m=\sum_{B\in\pi}|B|\geq 2b,$$

so

$$b\leq\frac{m}{2}.$$

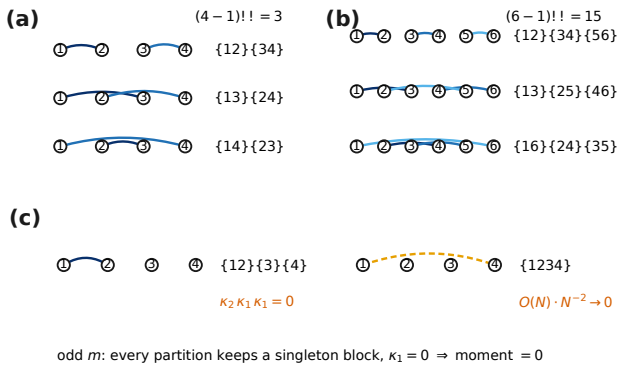


FIG. S2. Moment method. (a) Partitions containing singleton blocks are killed by $\kappa_1=0$, while a single large block contributes $O(N)/N^2\to 0$. (b) For $m=4$, the three perfect matchings survive and give σ^4. (c) For $m=6$, there are $(6-1)!!=15$ perfect matchings, each contributing σ^6. (d) For odd m, no perfect matching exists, so every partition contains a singleton and the moment vanishes.

Fix a partition π with b blocks. For each block B, fix one of its indices, say the smallest element. Because the cumulant decays exponentially in the diameter of the block by (A3), the sum over the remaining indices in B is absolutely convergent and uniformly bounded in N. Therefore each block contributes a factor of order $O(N)$ from the single free index. Thus the product over blocks contributes

$$O(N^b).$$

Multiplying by the prefactor $N^{-m/2}$, the contribution of π to $M_m(N)$ is

$$O(N^{b-m/2}).$$

If $b<m/2$, this contribution tends to zero as $N\to\infty$. Therefore only partitions with $b=m/2$ can survive. Since b is an integer, this is possible only when m is even, and it forces every block to have size exactly 2. Such partitions are precisely the perfect matchings of $\{1,\dots,m\}$.

Now let m be even and let π be a perfect matching. It consists of $m/2$ pairs. For one pair (a,b), the corresponding sum is

$$\sum_{i,j\in\Lambda}\kappa_2(X_i,X_j).$$

By translation invariance,

$$\kappa_2(X_i,X_j)=\kappa_2(X_0,X_{j-i}),$$

so

$$\sum_{i,j\in\Lambda}\kappa_2(X_i,X_j)=N\sum_{j\in\Lambda}\kappa_2(X_0,X_j)=N\,\sigma_N^2,$$

where we define

$$\sigma_N^2:=\sum_{j\in\Lambda}\kappa_2(X_0,X_j).$$

By (A3) the series is absolutely convergent uniformly in N, so

$$\sigma^2:=\lim_{N\to\infty}\sigma_N^2$$

exists and is finite. Moreover,

$$\sigma^2=\lim_{N\to\infty}\frac{1}{N}\tau(A_N^2)\geq 0,$$

so σ^2 is non-negative.

Thus each perfect matching contributes

$$N^{m/2}\,(\sigma_N^2)^{m/2}$$

before multiplying by $N^{-m/2}$. After the prefactor, the contribution becomes

$$(\sigma_N^2)^{m/2}\xrightarrow[N\to\infty]{}\sigma^m.$$

The number of perfect matchings of m elements is

$$(m-1)!!=(m-1)(m-3)\cdots 1.$$

Therefore

$$\lim_{N\to\infty}M_m(N)=\begin{cases}(m-1)!!\,\sigma^m, & m \text{ even},\\ 0, & m \text{ odd}.\end{cases}$$

These are exactly the moments of a centred Gaussian random variable with variance σ^2. Since the Gaussian distribution is uniquely determined by its moments, the empirical spectral distribution of $A_N/\sqrt{N}$ converges weakly to $\mathcal{N}(0,\sigma^2)$. □

Conserved sectors: hypotheses, remarks and proofs

In this appendix we extend the discussion to a common setting in soft-matter physics, where the system is subject to a symmetry or a conservation law. Rather than invoking Noether's theorem, we characterise a conservation law by an operator Q that commutes with the global operator A_N, in analogy with the Hamiltonian and number operator in quantum mechanics. Under the same exponential decay conditions, of the Lieb–Robinson type used in the main text, we obtain a Gaussian distribution for the eigenvalues of the (normalised) operator. However, the precise structure beyond these assumptions in practice often requires careful attention; we defer a detailed discussion of examples and counterexamples to a later section.

Conserved quantity and sector decomposition

Let Q representing the desired symmetry be a Hermitian operator on $\mathcal{H}$ such that

$$[Q, A_N] = 0.$$

Write its spectral decomposition, with P_q the orthogonal projection onto the eigenspace

$$Q = \sum_q q\, P_q, \qquad \mathcal{H}_q := P_q \mathcal{H}.$$

Since A_N commutes with Q, each $\mathcal{H}_q$ is invariant under A_N. We denote by $\mathcal{H}_q := \ker(Q - qI)$ the eigenspace of Q with eigenvalue q, and by P_q the orthogonal projection onto $\mathcal{H}_q$. We denote the sector as $Q = q$, then the normalized trace τ_q and the probability p_q of sector q as:

$$\tau_q(\cdot) := \frac{1}{\dim \mathcal{H}_q} \operatorname{tr}_{\mathcal{H}_q}(\cdot), \qquad p_q := \frac{\dim \mathcal{H}_q}{2^N}.$$

Such decomposition allows us to discuss and verify the condition for a Gaussian spectrum independently in each sector, and conditions for the overall Gaussian spectrum relies on such restriction to be consistent on all sectors of Q.

Hypotheses

We impose the following conditions, which are the natural analogues of (A2)–(A3) in each sector, together with a concentration requirement.

(C0) *Translation invariance of the conserved sectors.* We assume that the conserved charge Q commutes with all lattice translation operators:

$$[Q, U_k] = 0 \qquad \forall\, k = 1, \dots, d.$$

Equivalently, each eigenspace $\mathcal{H}_q$ is invariant under translations. Consequently, the sector trace τ_q satisfies

$$\tau_q(X_{i+a}) = \tau_q(X_i) \qquad \forall\, a, i \in \Lambda,$$

and, more generally, the sector cumulants are translation invariant:

$$\kappa_m^{(q)}(X_{i_1+a}, \dots, X_{i_m+a}) = \kappa_m^{(q)}(X_{i_1}, \dots, X_{i_m}) \qquad \forall\, a \in \Lambda.$$

(C1) *Zero mean in typical sectors.* For all q with $p_q > 0$,

$$\tau_q(X_i) = 0.$$

(In practice, a less strict condition is proposed in Appendix .)

(C2) *Exponential clustering of sector cumulants.* For each q and each ordered m-tuple $(i_1, \dots, i_m)$, define the sector cumulants $\kappa_m^{(q)}(X_{i_1}, \dots, X_{i_m})$ recursively by the same formula as in (A3), but with τ replaced by τ_q. We assume that there exist constants $C_m > 0$ and $\mu > 0$, independent of q and N, such that for all $m \geq 2$,

$$\left|\kappa_m^{(q)}(X_{i_1}, \dots, X_{i_m})\right| \leq C_m\, e^{-\mu\, \operatorname{diam}(i_1, \dots, i_m)}.$$

(C3) *Concentration of the sector variance.* For each q, define

$$\sigma_q^2 := \sum_{j \in \Lambda} \kappa_2^{(q)}(X_0, X_j).$$

We assume that σ_q^2 is finite uniformly in N and that, with respect to the distribution $\{p_q\}$,

$$\operatorname{Var}_Q\left(\sigma_Q^2\right) \xrightarrow[N\to\infty]{} 0.$$

Equivalently, the sector variance is essentially constant on typical sectors.

(C4) *Exponential decay of sector classical cumulants of centred row-sum functions.* For each sector q, let

$$T_i^q(x) := S_i(x) - \mathbb{E}_q[S_i],$$

where $\mathbb{E}_q$ is the uniform average over basis states in $\mathcal{H}_q$. Define the classical cumulants $\kappa_m^{\mathrm{cl},q}(T_{i_1}^q, \dots, T_{i_m}^q)$ with respect to $\mathbb{E}_q$ as in (C2), with the operator variables replaced by the functions T_i^q. We assume that there exist $C_m > 0$ and $\mu > 0$, independent of q and N, such that for all $m \geq 2$,

$$\left|\kappa_m^{\mathrm{cl},q}(T_{i_1}^q, \dots, T_{i_m}^q)\right| \leq C_m\, e^{-\mu\, \operatorname{diam}(i_1, \dots, i_m)}.$$

Theorem 2 is stated in the main text, where its hypotheses (C0)–(C3) are listed above; its proof is Appendix .

Remark 2. Assumption (C0) is a natural restriction in our framework. It ensures translational invariant conserved charge and is essential to valid the CLT argument. In the presence of a strong centre or a point source that breaks translation invariance, the basic premise of the CLT summation would be destroyed. Additionally, for system of negligible edge effects, periodic boundary condition is consistent with (C0) and enables the theoretical analysis. In the thermodynamic limit $N \to \infty$, boundary effects are negligible in any case, so (C0) is both physically intuitive and technically convenient.

Assumption (C1) requires the sector trace τ_q to have zero mean, which is the sector analogue of (A2). While (A2) is global, (C1) imposes the same condition on each conserved sector. In general, the sector mean may be non-zero even if the global mean vanishes, and the precise relation between the sector means and the global mean is discussed in Appendix , where we show that a non-zero sector mean can be absorbed by a phase shift in the limiting distribution without altering its Gaussian nature, provided the shift is of order $1/\sqrt{N}$.

The exponential decay of sector cumulants in (C2) is the direct analogue of (A3) within each conserved sector. It is the key input that ensures the independence of distant sites even after conditioning on the value of the conserved charge. Similarly, assumption (C4) is the sector version of (A5) for the classical row-sum functions: it imposes exponential decay on the classical cumulants of the centred row sums in each sector, in complete analogy with the relationship between (A3) and (A5). Thus the pair (C2) and (C4) mirrors the pair (A3) and (A5), respectively, the former controlling the operator cumulants and the latter controlling the classical row-sum cumulants.

Remark 3. Assumption (C3) is not an automatic consequence of a conserved charge. Instead, it addresses the concentration of the sector variance σ_q^2. As established in the proof of Theorem 2, conditions (C0)–(C2) already guarantee that, within each fixed sector q, the conditional distribution of $A_N/\sqrt{N}$ converges to a centred Gaussian with variance σ_q^2. However, the global (unconditional) spectral distribution is a mixture of these sector Gaussians weighted by p_q. Such a mixture is not generally Gaussian unless the variances σ_q^2 are essentially constant across the sectors that carry significant weight. Assumption (C3) ensures precisely the condition of trivial symmetry, because it requires the variance of σ_Q^2 with respect to the sector distribution vanishes in the thermodynamic limit, and the mixture collapses to a single Gaussian with variance $\sigma^2 = \lim_{N\to\infty} \mathbb{E}_Q[\sigma_Q^2]$.

To see this explicitly, note that the sector decomposition induced by the conserved quantity Q is orthogonal:

$$\mathcal{H} = \bigoplus_q \mathcal{H}_q, \qquad P_q P_{q'} = \delta_{q,q'} P_q.$$

Since $[Q, A_N] = 0$, the adjacency matrix is block diagonal with respect to this decomposition:

$$A_N = \bigoplus_q A_N^{(q)}, \qquad A_N^{(q)} := P_q A_N P_q.$$

Consequently, the full empirical spectral density is the weighted mixture

$$\rho_{A_N}(\lambda) = \sum_q p_q \, \rho_{A_N^{(q)}}(\lambda), \qquad p_q = \frac{\dim \mathcal{H}_q}{2^N}.$$

Thus, even if each sector converges to a Gaussian, the global limit is a Gaussian mixture; the condition (C3) is precisely what forces that mixture to be a single Gaussian.

Remark 4. Suppose now that, as in the theorem, in each sector q the spectral density of $A_N^{(q)}/\sqrt{N}$ converges to a centered Gaussian with variance σ_q^2. Then the characteristic function of the full normalized adjacency matrix is, in the limit,

$$\Phi(t) = \lim_{N\to\infty} \tau\Big(e^{itA_N/\sqrt{N}}\Big) = \sum_q p_q \, e^{-t^2\sigma_q^2/2} = \mathbb{E}_Q\Big[e^{-t^2\sigma_Q^2/2}\Big].$$

If we require the full limiting distribution to be exactly Gaussian with variance σ^2, then for all t,

$$\mathbb{E}_Q\Big[e^{-t^2\sigma_Q^2/2}\Big] = e^{-t^2\sigma^2/2}.$$

By Jensen's inequality applied to the convex function $x \mapsto e^{-t^2 x/2}$ for $t \neq 0$, we have

$$\mathbb{E}_Q\Big[e^{-t^2\sigma_Q^2/2}\Big] \geq e^{-t^2\,\mathbb{E}_Q[\sigma_Q^2]/2}.$$

Equality holds if and only if σ_Q^2 is almost surely constant and equal to $\mathbb{E}_Q[\sigma_Q^2]$. Therefore, if the full limit is a single Gaussian, then necessarily

$$\sigma_Q^2 = \sigma^2 \quad \text{almost surely with respect to } \{p_q\}.$$

In other words, the concentration condition (C3) is not merely a sufficient bound, it is forced by the requirement that a mixture of sector-wise Gaussians again be Gaussian. Thus, any attempt to replace (C3) by a more general functional relation $q \mapsto \sigma_q^2$ will, in the exact limit, either violate overall Gaussian or reduce to the constant-variance case. If one allows only approximate Gaussian, then small fluctuations of σ_q^2 around a typical value produce a scale mixture of Gaussians, which is generally not Gaussian but may still have Gaussian-like tails. A more general analysis could therefore consider such scale mixtures or allow non-zero sector means, but those lie outside the strict Gaussian framework considered here.

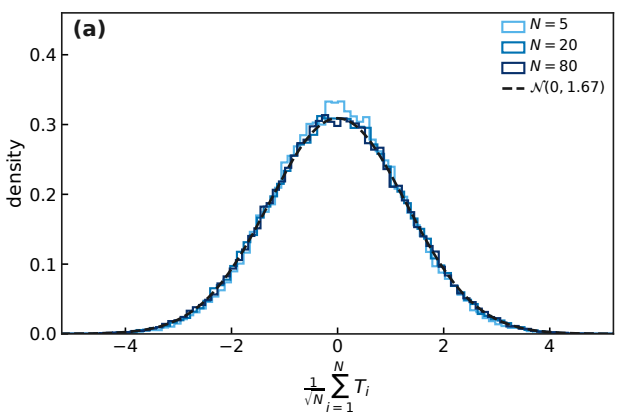


FIG. S3. Connectivity fluctuations in a toy model. Each local centred row-sum contribution T_i is modelled as a stationary AR(1) sequence with unit marginal variance and correlation $\rho = 0.25$, so that distant contributions are weakly dependent. For this toy model the limiting row-sum variance is $\sigma^2_{\text{row}} = (1+\rho)/(1-\rho) = 5/3 \approx 1.667$. The normalized total row sum $N^{-1/2}\sum_{i=1}^{N} T_i$ is sampled with 60,000 realizations and histogrammed with 90 bins for $N = 5, 20, 80$. As N increases, the empirical distribution approaches $\mathcal{N}(0, \sigma^2_{\text{row}})$ (black dashed line), in agreement with Proposition 1 on Gaussian connectivity fluctuations.

In order to understand the geometry of the configuration network of the phase space, we need to focus on the degree (also called connectivity) of each vertex, which is defined as the number of edges incident to that vertex. For the symmetric adjacency matrix A_N, the degree of a configuration vertex is precisely its row sum (equivalently, column sum). In the hypercube model, every vertex has the same degree N, which is the consequence of the completely commutative operators X_j. For general interacting systems, the row sums become site-dependent random variables, and their correlation are controlled by the classical cumulant decay in assumption (A5). With further restrictions in the form of the Lieb-Robinson bounds on the row-sum functions, one can deduce the asymptotic distribution of the normalised row sums as $N \to \infty$. The proposition stated in the main text formalises this result.

Its proof is Appendix .

Proof of Theorem 2

Proof. Fix a sector q. By (C1)–(C2), the assumptions of Theorem 1 are satisfied in the Hilbert space $\mathcal{H}_q$ with trace τ_q. Therefore, for every integer $m \geq 0$,

$$\lim_{N\to\infty} \tau_q\left[\left(\frac{A_N}{\sqrt{N}}\right)^m\right] = \begin{cases} (m-1)!!\,\sigma_q^m, & m \text{ even}, \\ 0, & m \text{ odd}. \end{cases}$$

Hence the moments of $\frac{A_N}{\sqrt{N}}$ in sector q converge to those of $\mathcal{N}(0, \sigma_q^2)$.

Now the full characteristic function of $\frac{A_N}{\sqrt{N}}$ is

$$\Phi_N(t) := \tau\left(e^{it\frac{A_N}{\sqrt{N}}}\right) = \sum_q p_q\, \tau_q\left(e^{it\frac{A_N}{\sqrt{N}}}\right).$$

For fixed t, the previous moment convergence implies

$$\lim_{N\to\infty} \tau_q\left(e^{it\frac{A_N}{\sqrt{N}}}\right) = e^{-t^2\sigma_q^2/2}$$

uniformly on compact sets of q under (C3). Since σ_q^2 concentrates around σ^2, we get

$$\Phi_N(t) \xrightarrow[N\to\infty]{} e^{-t^2\sigma^2/2}.$$

By Lévy's continuity theorem, the spectral distribution converges weakly to $\mathcal{N}(0, \sigma^2)$. □

Proof of Proposition 1

Proof. The proof is identical to that of Theorem 1 and Theorem 2 after the following replacements:

- replace the non-commuting operators X_i by the classical local functions T_i (or T_i^q in the sector case);
- replace the trace τ by the uniform expectation $\mathbb{E}_x$ (or $\mathbb{E}_q$);
- replace assumption (A3) by (A5) (or (C2) by (C4) in the sector case).

The cumulant expansion, the counting of partitions, and the pairing dominance argument are unchanged. Therefore the moment convergence and the Gaussian limit follow exactly as in the proof of Theorem 1 and Theorem 2. □

Decimation renormalisation

The decimation map

To begin with, we discuss whether the present model admits a decimation-type renormalization mapping analogous to that of the Ising model. The short answer is affirmative in the asymptotic sense, provided the locality and clustering assumptions introduced earlier are satisfied. However, the non-commuting nature of the X_i forces the mapping to be understood at the level of operator-valued cumulants rather than as a scalar recursion. Partition the lattice Λ into disjoint blocks $\{B_\alpha\}$ of linear size ℓ, and define the block operator

$$Y_\alpha := \sum_{i\in B_\alpha} X_i. \tag{8}$$

Then $A_N = \sum_\alpha Y_\alpha$. By translation covariance (A1), the Y_α form an approximately translation-invariant family on the coarse-grained lattice. The exponential decay of cumulants (A3) is preserved under block summation. Indeed, for any ordered tuple of blocks $(\alpha_1, \dots, \alpha_m)$,

$$\kappa_m(Y_{\alpha_1}, \dots, Y_{\alpha_m}) = \sum_{i_1 \in B_{\alpha_1}} \cdots \sum_{i_m \in B_{\alpha_m}} \kappa_m(X_{i_1}, \dots, X_{i_m}).$$

Using (A3) and summing over the block volumes, we obtain

$$\begin{aligned} &|\kappa_m(Y_{\alpha_1}, \dots, Y_{\alpha_m})| \\ &\le C_m \, |\ell|^m \, e^{-\mu \, \mathrm{dist}(\alpha_1, \dots, \alpha_m)}, \end{aligned} \tag{9}$$

where dist is the distance between blocks. Hence, after rescaling the cumulant constants, the coarse-grained variables Y_α satisfy the same Lieb-Robinson type exponential clustering as the original X_i. Similarly, the quasi-locality bound (A4) remains valid for the block variables, with a modified decay rate.

Formally, we fix ℓ and define the coarse-grained system on the quotient lattice Λ/ℓ with operators Y_α. By the stability of the cumulant bounds, the normalized sum $A_N/\sqrt{N}$ has the same Gaussian limit as the original model. Thus the mapping

$$\mathcal{R}_\ell : \quad \{X_i\}_{i \in \Lambda} \longmapsto \{Y_\alpha\}_{\alpha \in \Lambda/\ell}$$

is a renormalization group (RG) transformation that preserves the Gaussian fixed point (GFP). This is the analogue of the Ising decimation map. However, whereas in the Ising model the transformation acts on scalar coupling constants, here it acts on the space of local operator families satisfying (A1)–(A3). The fixed point is characterized by the Gaussian moment relations

$$\kappa_m = 0 \quad \text{for } m \ge 3, \qquad \kappa_4 = 0, \qquad \mu_4 = 3\mu_2^2,$$

where μ_m denotes the m-th moment of the limiting distribution.

If a conserved charge Q is present and sector assumptions (C1)–(C4) hold, the same construction applies within each sector $\mathcal{H}_q$. The coarse-grained variables $Y_\alpha^q := \sum_{i \in B_\alpha} X_i$ remain well-defined on $\mathcal{H}_q$, and the sector cumulant decay (C2) is stable under block summation in exactly the same way, yielding an analogue of (9) with τ_q. Consequently, the decimation mapping preserves the Gaussian limit in every sector, and after averaging over sectors the global limit remains Gaussian.

The main difference from the classical Ising case is that $\mathcal{R}_\ell$ is not exact at finite ℓ. Since the X_i do not commute, the block variables Y_α generate effective multi-operator corrections via nested commutators. These corrections are controlled by (A4) and are exponentially small in the block separation, hence the decimation map is an asymptotic RG transformation, valid in the limit of large separation, rather than an exact algebraic identity. In models with all-to-all interactions, such as the Johnson model, the cumulant decay fails, the block variables acquire long-range correlations, and the Gaussian fixed point is no longer stable.

Row-sum covariance under decimation

We now examine the distribution of connectivity under decimation. With the same block partition, define the centred block row-sum function

$$U_\alpha(x) := \sum_{i \in B_\alpha} T_i(x), \tag{10}$$

where T_i is the centred local row-sum from (A5). Then the total centred row sum is $\widetilde{R}_N(x) = \sum_\alpha U_\alpha(x)$. The classical cumulants of the block variables U_α satisfy

$$\kappa_m^{\mathrm{cl}}(U_{\alpha_1}, \dots, U_{\alpha_m}) = \sum_{i_1 \in B_{\alpha_1}} \cdots \sum_{i_m \in B_{\alpha_m}} \kappa_m^{\mathrm{cl}}(T_{i_1}, \dots, T_{i_m}),$$

and using (A5) we obtain

$$\begin{aligned} &\left|\kappa_m^{\mathrm{cl}}(U_{\alpha_1}, \dots, U_{\alpha_m})\right| \\ &\le C_m \, |\ell|^m \, e^{-\mu \, \mathrm{dist}(\alpha_1, \dots, \alpha_m)}. \end{aligned} \tag{11}$$

Thus the block functions U_α satisfy the same exponential clustering as the original T_i, with rescaled constants. By Proposition 1, the normalized block row sum

$$\frac{1}{\sqrt{|\Lambda/\ell|}} \sum_\alpha U_\alpha(x)$$

converges weakly to a centred Gaussian. Since $|\Lambda/\ell| = N/\ell^2$, we have

$$\frac{1}{\sqrt{|\Lambda/\ell|}} \sum_\alpha U_\alpha(x) = \ell \, \frac{\widetilde{R}_N(x)}{\sqrt{N}},$$

so the decimated connectivity has a distribution equivalent to that of the original one up to the deterministic factor ℓ. The limiting variance scales as $\ell^2 \, \sigma_{\mathrm{row}}^2$. In the presence of a conserved charge, the sector block functions U_α^q satisfy an analogous bound to (11) with (C4), and the same conclusion holds within each sector.

It is worth to notice that, while the details of the highly complex network of configurations is an extremely challenging question, we have the capability to detect (asymptotic) self-similar structures upon RG decimation transformation. In other words, by coarse-graining the physical space and apply decimation, the distributions

discussed here stay invariant asymptotically upon rescaling, and we deduce that the network is therefore self-similar. Han et al has investigated examples with such phenomena and named this self similarity as "fractal network". In next section we shall discuss the typical examples they proposed and simulated, with brief discussion on verify the LR bound.

Approximate conservation and spectral shift

We now consider the situation where the conservation law is only approximate: $[Q, A_N] \neq 0$. We show that under a mild and physically natural condition, the Gaussian limit is preserved up to a possible shift.

Locality of the conserved quantity

Throughout this appendix we assume that Q is a sum of local terms,

$$Q = \sum_{\boldsymbol{x}\in\Lambda} q_{\boldsymbol{x}},$$

where each $q_{\boldsymbol{x}}$ is supported on a finite region around $\boldsymbol{x}$ and has integer spectrum. Typical examples are total particle number $N = \sum_{\boldsymbol{x}} n_{\boldsymbol{x}}$ or total magnetization $S^z = \sum_{\boldsymbol{x}} S^z_{\boldsymbol{x}}$. In particular, the spectral gap of Q satisfies

$$\Delta_Q \geq 1,$$

uniformly in the system size N. This fact is essential because it prevents the energy denominators in perturbation theory from becoming arbitrarily small as $N \to \infty$.

Define the block-diagonal part of A_N with respect to Q:

$$A_{\mathrm{D}} := \sum_q P_q A_N P_q,$$

where P_q is the projection onto the eigenspace of Q with eigenvalue q. The off-diagonal part is

$$A_{\mathrm{OD}} := A_N - A_{\mathrm{D}}.$$

By construction $[A_{\mathrm{D}}, Q] = 0$, so A_{D} falls within the setting of the main theorem, provided the sector cumulant assumptions (C1)–(C3) hold for A_{D}.

Gaussian limit with a possible shift

We assume that the commutator is small in the following local sense:

$$\|[Q, A_N]\| = o(\sqrt{N}),$$

and that $[Q, A_N]$ is a sum of local terms with exponentially decaying tails. The second condition is automatic if both Q and A_N are built from local operators with finite range, by the Lieb–Robinson bounds already used in the main text.

Remark 5. Since Q has spectral gap $\Delta_Q \geq 1$ and A_{OD} is the off-diagonal part with respect to Q, standard perturbation theory gives

$$\|A_{\mathrm{OD}}\| \leq C\,\frac{\|[Q, A_N]\|}{\Delta_Q} \leq C\,\|[Q, A_N]\| = o(\sqrt{N}).$$

Hence

$$\frac{\|A_{\mathrm{OD}}\|}{\sqrt{N}} \to 0.$$

In particular, for every fixed integer $m \geq 1$,

$$|\tau(A_{\mathrm{OD}}^m)| \leq \|A_{\mathrm{OD}}\|^m = o(N^{m/2}).$$

Since $\|A_{\mathrm{OD}}/\sqrt{N}\| \to 0$, the spectral distribution of

$$\frac{A_N}{\sqrt{N}} = \frac{A_{\mathrm{D}}}{\sqrt{N}} + \frac{A_{\mathrm{OD}}}{\sqrt{N}}$$

is asymptotically the same as that of $A_{\mathrm{D}}/\sqrt{N}$. Under assumptions (C1)–(C3) for A_{D} in each sector, the main theorem implies that $A_{\mathrm{D}}/\sqrt{N}$ converges weakly to a Gaussian law with variance σ^2. Therefore $A_N/\sqrt{N}$ also converges to a Gaussian law, possibly with a shifted mean given by

$$\mu = \lim_{N\to\infty} \frac{\tau(A_{\mathrm{OD}})}{\sqrt{N}},$$

provided this limit exists. If $\tau(A_{\mathrm{OD}}) = o(\sqrt{N})$, the shift is zero.

The present setting is physically relevant when the violation of conservation is weak and local, e.g. due to a small non–number-conserving perturbation of finite range. In models with all-to-all couplings, such as the Johnson model, the commutator $\|[Q, A_N]\|$ is typically of order $O(N)$ rather than $o(\sqrt{N})$, and the sector cumulants do not decay with distance; consequently the Gaussian limit fails. This shows that locality of both the operators and the conservation-breaking terms is essential.

Worked examples

This appendix reproduces the discussion of the two models that motivated the analysis; the corresponding figure is Fig. 1 of the main text.

We now illustrate how the general conditions developed in the main text are satisfied in two canonical models, i.e. the frustrated spin glass and the lattice gas. Both have been studied numerically for small system sizes in

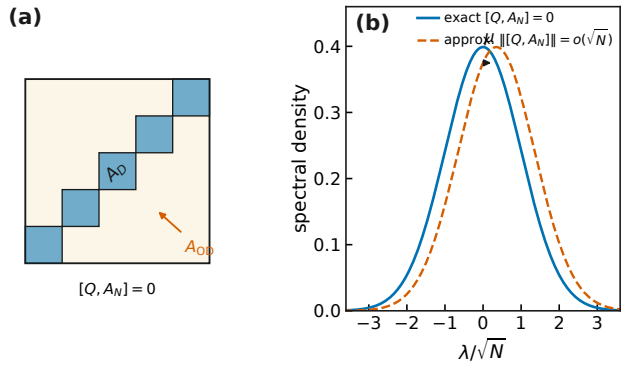


FIG. S4. Approximate conservation and spectral shift. (a) The global operator decomposes as $A_N = A_D + A_{\mathrm{OD}}$, where A_D is block diagonal with respect to the eigenspaces of Q, and A_{OD} is the off-diagonal part. (b) If the commutator is small in the local sense $\|[Q, A_N]\| = o(\sqrt{N})$, the Gaussian limit survives up to a possible shift $\mu = \lim_{N\to\infty} \tau(A_{\mathrm{OD}})/\sqrt{N}$. (c) The off-diagonal part is subleading: $\|A_{\mathrm{OD}}\|/\sqrt{N} \to 0$ as $N \to \infty$.

the work of Han et al. (2015), where the emergence of Gaussian spectral densities was observed as N increases. Our aim here is not to perform new computations, but rather to indicate why these models naturally fall within the framework of our assumptions, and to highlight the sense in which the verification is essentially trivial once the local structure is identified.

Consider first the frustrated spin glass. The Hamiltonian consists only of nearest-neighbour interactions on a lattice, with frustration arising from competing couplings. Since all interactions are strictly local and of finite range, the associated local operators X_i have finite support. Consequently, the multiple commutators in assumption (A4) vanish identically for disjoint supports, and the exponential decay of cumulated traces in (A3) follows immediately from the product structure of the maximally mixed state. Likewise, the classical cumulant decay in (A5) is automatic because the row-sum functions are built from these same local terms. Thus, apart from the trivial shift of the mean (which can be removed by centring), the spin glass satisfies all assumptions of Theorem 1 and Theorem 2 without any further input.

For the lattice gas model with a conserved particle number, the situation is analogous. The conserved charge Q is the total particle number operator, and the local terms X_i describe hopping processes that exchange occupied and empty sites on neighbouring lattice sites, where each X_i acts only on a pair of adjacent vertices. This dynamics is manifestly local, hence the sector decomposition induced by Q inherits the same finite support property, and the sector cumulants in (C2) as well as the classical sector cumulants in (C4) decay exponentially for the same reason. Thus the lattice gas with fixed number of particles satisfies all the necessary hypotheses (except for (C3)) of Theorem 2, and the global Gaussian limit follows.

In both examples, the clustering assumptions reduces to the finite range interactions. There is therefore no need to invoke more elaborate estimates since the Lieb Robinson bounds are automatically satisfied with exponential tails, and the cumulant decay constants are uniform in the system size. This explains why the Gaussian universality observed numerically is robust and does not depend on fine details of the model.

Finally, we note that the discussion above has been phrased for a simple cubic lattice for concreteness. For closed-packed structures such as the face-centred cubic (FCC) lattice, one may regard the lattice as a motif convoluted with a simple cubic lattice; the locality and translation covariance properties remain unchanged. For hexagonal close-packed (HCP) structures in three or higher dimensions, the generator of translations is slightly more involved, but the generalisation is straightforward and does not alter the validity of the assumptions. The key requirement of finite range interactions is unaffected by the underlying Bravais lattice.